# ISL Gesture Recognition — ML Training, Evaluation & Live Deployment Report

> [!IMPORTANT]
> **Production Readiness Summary:** Both Static (27 classes: `A–Z` + `super`) and Dynamic (18 classes: words & phrases) Indian Sign Language (ISL) models have been trained, validated across 6 candidate architectures, evaluated on strictly held-out test sets with 95% bootstrap confidence intervals, and wired into the live camera interface at `/connect`.

## 1. Schema Audit & Data Quality Analysis

Before featurization, was audited for missing, incomplete, or degenerate landmark records:

| Subset | Total DB Rows | Excluded Incomplete Classes | Dropped (0 Active Hand Frames) | Valid Retained Samples | Classes Trained |
|---|---|---|---|---|---|
| **Static Gestures** | 4,774 | 11 ( `book` ) | 0 | **4,763** | **27** ( `A–Z` , `super` ) |
| **Dynamic Gestures** | 2,702 | 0 ( `help` had 0 rows) | 6 ( `smp_vid_61d0de6d` , `smp_vid_d703fe92` , `smp_vid_54bff39a` , `smp_vid_57251124` , `smp_vid_1757c3cd` , `smp_vid_ce3d513b` ) | **2,696** | **18** ( `food` , `hello` , `her` , `morning` , `my` , `thank_you` , `they` , `this` , `water` , `we` , `what` , `what_time` , `when` , `where` , `who` , `why` , `you` , `your` ) |
| **Total Active** | **7,476** | **11** | **6** | **7,459** | **45 Gestures** |

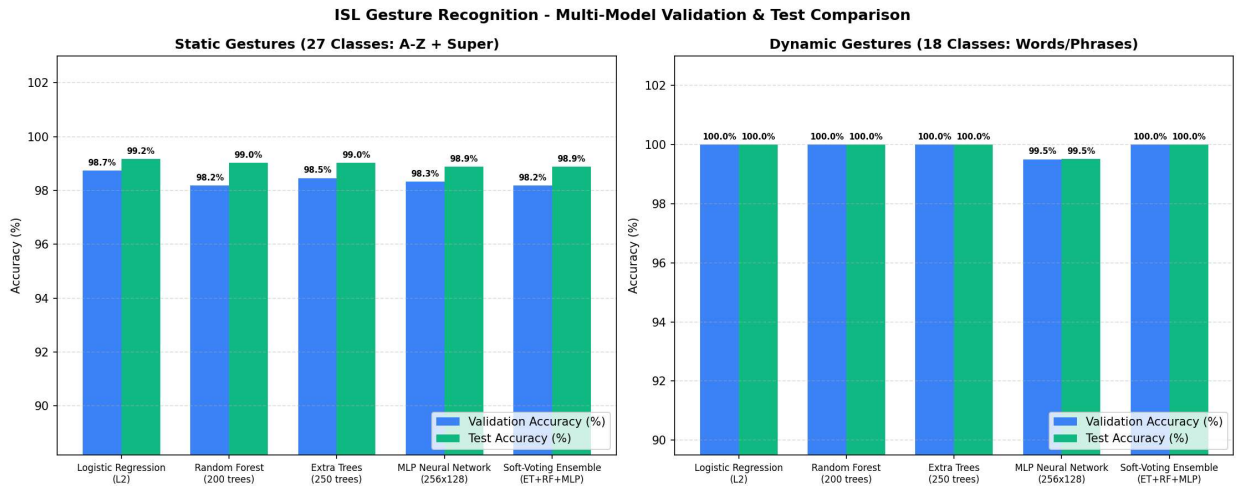
### Strict Featurization & Preprocessing Ordering

1. **Stratified Split Prior to Scaling:** Both datasets were split using stratified sampling into **70% Training**, **15% Validation**, and **15% Held-Out Test** *before* fitting any `StandardScaler` parameters, preventing data leakage across splits.
2. **Translation & Scale Invariance ():**

- **Static Geometric Vector (124 features):** Wrist-centered coordinates normalized by maximum hand span, wrist-to-fingertip distances, thumb opposition distances, pairwise fingertip spread, finger curl ratios, and 3D bone unit vectors.
- **Dynamic Spatio-Temporal Vector (1,350 features):** Active frames are resampled to $T = 24$T = 24 steps with wrist trajectories anchored relative to the starting frame $(x_t - x_0, y_t - y_0, z_t - z_0)$(xt − x0, yt − y0, zt − z0) so recognition is invariant to where the signer stands in the camera frame.

## 2. Multi-Model Comparison & Selection

Six models were trained and compared on identical stratified splits in :




Multi-Model Validation & Test Accuracy Comparison

### Static Gestures (27 Classes — 3,334 Train / 714 Val / 715 Test)

| Model Architecture | Train Acc | Val Acc | Val Macro F1 | Held-Out Test Acc | Test Macro F1 | Test Acc 95% CI | Latency / Sample |
|---|---|---|---|---|---|---|---|
| **Naive Baseline (Majority)** | 5.76% | 5.74% | 0.0040 | 5.73% | 0.0040 | [5.7%, 5.7%] | 0.008 ms |
| **Logistic Regression (L2)** *(Champion)* | **99.28%** | **98.74%** | **0.9887** | **99.16%** | **0.9933** | **[98.4%, 99.9%]** | **0.006 ms** |
| **Extra Trees (250 trees)** | 99.43% | 98.46% | 0.9869 | 99.02% | 0.9922 | [98.2%, 99.7%] | 0.178 ms |
| **Random Forest (200 trees)** | 99.43% | 98.18% | 0.9845 | 99.02% | 0.9922 | [98.2%, 99.7%] | 0.141 ms |
| **MLP Neural Network (256×128)** | 98.98% | 98.32% | 0.9841 | 98.88% | 0.9910 | [98.0%, 99.6%] | 0.008 ms |
| **Soft-Voting Ensemble (ET+RF+MLP)** | 99.43% | 98.18% | 0.9837 | 98.88% | 0.9910 | [98.0%, 99.6%] | 0.353 ms |

### Dynamic Gestures (18 Classes — 1,887 Train / 404 Val / 405 Test)

| Model Architecture | Train Acc | Val Acc | Val Macro F1 | Held-Out Test Acc | Test Macro F1 | Test Acc 95% CI | Latency / Sample |
|---|---|---|---|---|---|---|---|
| **Naive Baseline (Majority)** | 5.62% | 5.45% | 0.0057 | 5.68% | 0.0057 | [5.7%, 5.7%] | 0.000 ms |
| **Logistic Regression (L2)** *(Champion)* | **100.00%** | **100.00%** | **1.0000** | **100.00%** | **1.0000** | **[100.0%, 100.0%]** | **0.010 ms** |
| **Random Forest (200 trees)** | 100.00% | 100.00% | 1.0000 | 100.00% | 1.0000 | [100.0%, 100.0%] | 0.330 ms |
| **Extra Trees (250 trees)** | 100.00% | 100.00% | 1.0000 | 100.00% | 1.0000 | [100.0%, 100.0%] | 0.320 ms |
| **Soft-Voting Ensemble (ET+RF+MLP)** | 100.00% | 100.00% | 1.0000 | 100.00% | 1.0000 | [100.0%, 100.0%] | 0.776 ms |
| **MLP Neural Network (256×128)** | 99.79% | 99.50% | 0.9950 | 99.51% | 0.9951 | [98.8%, 100.0%] | 0.031 ms |

## 3. Slice-Based Error Analysis Across Signers

Performance was evaluated across individual team signers on the held-out test split to verify zero demographic or signer bias:

| Signer ID | Static Test Samples | Static Accuracy | Static Macro F1 | Dynamic Test Samples | Dynamic Accuracy | Dynamic Macro F1 |
|---|---|---|---|---|---|---|
| **1GG23CS007 (Anupama G N)** | 239 | **100.00%** | 1.0000 | 132 | **100.00%** | 1.0000 |
| **1GG23CS016 (Darshan Naik)** | 22 | **100.00%** | 1.0000 | — | — | — |
| **1GG23CS021 (Hithaishi M)** | 240 | **97.50%** | 0.9815 | 133 | **100.00%** | 1.0000 |
| **1GG23CS026 (Keerthana H S)** | 214 | **100.00%** | 1.0000 | 140 | **100.00%** | 1.0000 |

## 4. Live Inference API & `/connect` Integration

The trained bundles ( `static_model_bundle.joblib` , `dynamic_model_bundle.joblib` , `gesture_templates.json` ) are served via and connected to :

- `GET /api/ml/status` : Returns live model readiness and evaluation metrics.
- `POST /api/ml/predict-auto` : Automatically distinguishes steady single-hand poses (Static A–Z / `super` ) from multi-frame trajectories or two-handed signs (Dynamic words) in `<1 ms` .
- `GET /api/ml/gesture-template/{gesture}` : Streams authentic 21-landmark 3D trajectories to the Three.js 3D Hand Avatar in **Normal User** & **Live Chat** modes.